

Baddel: AI-Powered Smart Recycling System

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Abstract—This paper addresses the challenges in recycling systems, particularly the issue of inefficient sorting and classification of materials. Traditional methods rely on manual processes, which are slow and prone to human error. To improve this, the study explores the use of deep learning techniques, specifically a YOLO-based object detection model. The dataset used in this project consists of labeled images that were cleaned and prepared through a structured preprocessing process, including removing unmatched images, validating labels, and handling duplicates. The model was tested on sample images and demonstrated the ability to detect multiple objects in real time and classify them accurately. Although the current implementation focuses on object detection, the results show that the system can be extended to classify recyclable materials such as plastic, metal, and wood. This approach provides a strong foundation for developing more efficient and automated recycling systems.

I. INTRODUCTION

The increasing amount of global waste has made recycling a critical component of environmental sustainability. As industrial production continues to grow, the demand for materials such as plastic, metal, glass, and paper has significantly increased. However, these materials do not perform equally within recycling systems. Some materials can be reused multiple times without losing their quality, while others degrade after recycling, limiting their long-term usability. This creates a gap between material consumption and environmental sustainability.

Plastic is one of the most widely used materials due to its low cost, flexibility, and ease of production. Despite these advantages, plastic recycling remains inefficient and contributes significantly to environmental pollution [1], [2]. In contrast, materials such as metals and glass offer higher recycling efficiency because they can be reused multiple times without significant quality loss.

Understanding the differences between materials and their recycling performance is essential for improving recycling systems and reducing environmental impact. Therefore, this study focuses on analyzing different recycling materials and exploring solutions to enhance classification and sorting processes.

II. LITERATURE REVIEW

A. Recycling Materials and Their Characteristics

Recycling materials are essential for reducing environmental impact and supporting sustainable development. Materials are commonly classified into plastics, metals, glass, paper, organic materials, and wood. Each material differs in terms of physical properties, recyclability, environmental impact, and processing requirements. Previous studies highlight that recycling efficiency is influenced by sorting accuracy, contamination, and processing techniques [3], [4].

Plastic is one of the most widely used materials due to its low cost, flexibility, and lightweight structure. It is heavily used in packaging, healthcare, and consumer products. However, plastic recycling is complex because different types must be separated before processing. In addition, plastic degrades after each recycling cycle, which limits its reuse and reduces efficiency [5], [6]. Advanced approaches such as chemical recycling have been introduced, but they require high energy and cost, limiting their practical application [7], [8]. As a result, plastic remains one of the least sustainable materials despite its widespread use [9].

Metals such as aluminum and steel are highly efficient in recycling systems. They can be reused multiple times without losing their properties, making them highly sustainable. Recycling metals also consumes less energy compared to producing new materials, which reduces environmental impact [10]. However, efficient recycling requires proper collection and sorting systems.

Glass is also highly recyclable and can be reused without losing quality. Recycling glass reduces the need for raw materials and supports sustainability. However, the process requires sorting by color and involves high-temperature processing, which increases energy consumption [11].

Paper is widely used in packaging and printing and is considered environmentally friendly because it is biodegradable and renewable. Recycling paper helps reduce deforestation, but paper fibers weaken after multiple recycling cycles. In addition, contamination from liquids or food reduces recycling efficiency [4].

Organic materials such as food waste are treated differently. Instead of direct reuse, they are converted into energy sources

such as biogas. This process supports sustainability but requires specialized systems and proper waste separation [11].

Wood is a renewable material commonly used in construction and furniture. It contributes positively to sustainability because it stores carbon and is biodegradable. Wood recycling typically involves reuse or conversion into products such as particleboard or energy. However, treated or contaminated wood is difficult to recycle and requires careful handling.

Overall, the literature shows a clear imbalance between material usage and sustainability. Materials like plastic are widely used but inefficient in recycling, while materials like metal and glass provide higher recycling efficiency.

B. Technical Approaches in Recycling Systems

In addition to material-focused studies, recent research has explored technical solutions to improve recycling systems, particularly in material classification and sorting. Traditional recycling relies heavily on manual sorting, which is slow, labor-intensive, and prone to human error. This limitation reduces recycling efficiency and increases contamination rates.

To address this issue, computer vision and deep learning techniques have been introduced. Convolutional Neural Networks (CNNs) are commonly used for image classification tasks, as they can automatically extract visual features such as shape, texture, and color [12]. However, CNN-based classification models are limited because they cannot detect multiple objects in real-time.

Object detection models such as YOLO (You Only Look Once) provide a more advanced solution. YOLO is a real-time detection model that processes an entire image in a single pass, making it significantly faster than traditional models [13]. The model divides the image into a grid and predicts object locations and classes simultaneously, allowing it to detect multiple materials at once.

Recent versions such as YOLOv4 have improved both accuracy and speed, making them suitable for real-world applications [14]. These models can operate under different conditions, including varying lighting, object sizes, and backgrounds.

Several studies have applied YOLO in recycling systems. For example, Boragaonkar et al. [15] developed a smart waste classification system using YOLO, achieving high accuracy in detecting recyclable materials. Similarly, Nasir and Roslan [16] proposed a real-time waste classification system that improved sorting efficiency compared to manual methods.

Despite these advancements, challenges remain. Dataset quality is a major limitation, as many datasets lack diversity and do not represent real-world conditions. In addition, some materials, such as plastic and wood, have similar visual features, making classification more difficult. Another challenge is the computational cost required for real-time processing.

Overall, AI-based systems, especially YOLO, provide an effective solution for improving recycling systems by enabling accurate and automated material classification, reducing human error, and increasing efficiency.

C. Comparative Analysis of Recycling Materials

The following table summarizes the key differences between major recycling materials in terms of usage, efficiency, sustainability, and limitations.

TABLE I
COMPARISON OF COMMON RECYCLABLE MATERIALS

Material	Usage	Recycle Eff.	Sustainability	Cost	Advantages	Limitations
Plastic	Very High	Low	Low	Low	Lightweight, cheap, flexible	Difficult to sort, degrades after recycling, pollution
Metal	Medium	Very High	Very High	High	Durable, infinitely recyclable, efficient	Expensive collection and processing
Glass	Medium	High	High	Medium	Reusable without quality loss	Requires sorting, high-temp processing
Paper	High	Medium	Medium	Low	Biodegradable, renewable	Weakens after recycling, contamination sensitive
Organic	Low	Medium	High	Medium	Produces biogas, reduces waste	Needs separation and special systems
Wood	Medium	Medium	High	Medium	Renewable, stores carbon, reusable	Difficult if treated, proper handling needed

The table highlights a clear gap between usage and sustainability. Plastic has the highest usage but the lowest recycling efficiency, while metals and glass provide better long-term sustainability. This comparison emphasizes the need for improved classification and sorting systems to enhance recycling performance.

III. SUMMARY OF FINDINGS

The findings of this study reveal significant differences in recycling performance across materials, particularly in terms of efficiency, environmental impact, and long-term sustainability. A key observation is the clear mismatch between material usage and recycling effectiveness.

Plastic remains the most widely used material due to its low cost and versatility; however, it demonstrates the lowest recycling efficiency. This is mainly due to its complex composition, the need for separation of different plastic types, and degradation after multiple recycling cycles. As a result, plastic contributes heavily to environmental pollution despite its practical advantages.

In contrast, metals and glass show the highest recycling efficiency and sustainability. These materials can be reused multiple times without significant loss of quality, making them more suitable for circular economy systems. Additionally, their recycling processes are more stable and predictable, although they still depend on effective collection and sorting systems.

Paper, organic materials, and wood provide moderate sustainability outcomes. Paper is biodegradable and renewable, but its fibers weaken after repeated recycling cycles. Organic materials support sustainability through energy recovery processes such as biogas production, although they require specialized infrastructure. Wood also offers environmental benefits as a renewable and carbon-storing material, but its recyclability is limited when it is treated or contaminated.

Another important finding is that the main limitation in recycling systems is not only the material properties themselves, but also the accuracy of classification and sorting processes. Inefficient or incorrect sorting leads to contamination, reduces material quality, and lowers overall system performance.

Therefore, improving material identification and classification is essential for enhancing recycling efficiency. This highlights the importance of adopting automated and intelligent systems, such as AI-based detection models, to reduce human error and improve sorting accuracy.

Overall, the study emphasizes that achieving sustainable recycling systems requires both material-aware strategies and advanced technological solutions to address the gap between high material usage and low recycling performance.

IV. PROBLEM DEFINITION

Despite the development of recycling systems, waste management remains inefficient due to inaccurate material classification and poor sorting processes. Recyclable materials such as plastic, metal, and wood are often mixed together, leading to contamination and reducing the overall quality of recycling output [1], [4].

A major challenge is the reliance on manual sorting, which is time-consuming and prone to human error. Variations in material appearance, lighting conditions, and object shapes make accurate identification difficult, especially in large-scale environments. As a result, materials such as plastic, which are widely used, often have low recycling efficiency due to incorrect classification [6], [9].

Although automated solutions have been proposed, many are not suitable for real-time applications or require high computational resources. In addition, existing models often struggle with detecting multiple materials in complex environments, highlighting a gap between research solutions and practical implementation [7].

To address this gap, this project proposes an automated image-based system using the YOLO model. YOLO enables real-time detection and classification of multiple materials in a single image, improving sorting accuracy and reducing human error. This approach aims to enhance recycling efficiency and support more sustainable waste management systems.

V. ANALYSIS AND DISCUSSION

The analysis shows a clear mismatch between material usage and recycling efficiency. Plastic is the most widely used material, yet it has the lowest recycling efficiency due to its complex structure, need for separation, and degradation after reuse [6], [9]. This makes plastic a major contributor to environmental pollution despite its advantages in cost and flexibility.

In contrast, materials such as metal and glass demonstrate high recycling efficiency and long-term sustainability. These materials can be reused multiple times without significant loss of quality, making them more suitable for effective recycling systems. However, their performance still depends on proper sorting and collection processes.

Paper, organic materials, and wood provide moderate sustainability. Paper is biodegradable but weakens after multiple recycling cycles, while organic materials require specialized processing systems. Wood is also a renewable material, but its

recycling efficiency depends on proper handling and whether it is treated or contaminated.

A key issue identified in this analysis is inaccurate material classification. Incorrect sorting leads to contamination, which reduces recycling quality and overall system efficiency. This indicates that the main limitation lies in how materials are identified and processed rather than the materials themselves.

Therefore, improving classification accuracy is essential for enhancing recycling systems. Automated detection using models such as YOLO can support this by enabling accurate and real-time material identification, reducing human error, and improving sorting efficiency.

VI. PROJECT OBJECTIVE AND FOCUS

The objective of this research is to develop an automated system for detecting and classifying recyclable materials using a YOLO-based object detection model. The system aims to improve the accuracy of material identification and reduce errors in traditional recycling processes.

This project focuses on building an image-based classification system that can detect materials such as plastic, metal, and wood in real time. YOLO is selected as the main model due to its strong performance in object detection tasks, providing a good balance between accuracy and speed, as well as ease of implementation.

The dataset used in this project is collected from Kaggle and consists of labeled images representing different material categories. The data will be cleaned and prepared by removing corrupted images, resizing them to a fixed size, and organizing them into training, validation, and testing sets.

The system is trained to recognize materials based on visual features such as shape, texture, and color, and to produce outputs including object class, bounding boxes, and confidence scores. This allows the model to detect multiple materials within a single image, simulating real-world recycling conditions.

Overall, the project aims to reduce reliance on manual sorting, improve classification accuracy, and enhance the efficiency of recycling systems through the use of deep learning techniques.

VII. METHODOLOGY

A. Data Information

The dataset used in this project was obtained from Kaggle, which provides publicly available datasets for machine learning and computer vision tasks. The dataset consists of labeled images representing different object categories such as chairs and tables.

An image-based dataset was selected because it reflects real-world scenarios where objects are identified visually. This allows the system to simulate practical environments and supports the development of a real-time object detection model using YOLO.

The dataset is used to train and evaluate the model, enabling it to detect multiple objects within a single image.

B. Data Description

The dataset contains images with corresponding annotations in YOLO format. Each image includes one or more objects, where each object is labeled with its class and bounding box coordinates.

The images vary in background, lighting conditions, object size, and viewing angles. This variability helps the model learn to detect objects under different conditions and improves its ability to generalize to new data.

The dataset was divided into training, validation, and testing sets to ensure proper evaluation of the model.

C. Data Cleaning

The dataset was carefully cleaned and prepared to ensure high-quality input for training the YOLO model. This process focused on improving data consistency and removing errors.

The following steps were applied:

- **Removing Unmatched Images:** Images without corresponding annotation files were identified and removed to ensure consistency.
- **Label Validation:** All label files were checked to confirm they follow the YOLO format, including correct class IDs and normalized bounding box values.
- **Duplicate Detection:** Duplicate images were detected using hashing techniques and removed to avoid bias.
- **Low-Quality Image Handling:** Small or unclear images were reviewed. Unusable images were removed, while acceptable ones were kept.
- **Semantic Validation:** Bounding boxes were visually inspected to ensure that annotations correctly match the objects.

These steps ensured that the dataset is clean, accurate, and suitable for training.

D. Feature Engineering, Regularization, and Augmentation

In this project, feature extraction is handled automatically by the YOLO model, which learns visual patterns such as shape and structure directly from the images.

To improve model performance and reduce overfitting, the following techniques were applied:

Data Augmentation:

- Horizontal and vertical flipping
- Rotation and scaling
- Brightness and contrast adjustments

These techniques increase dataset diversity and help the model perform better under different conditions.

Regularization:

- Early stopping during training
- Weight decay
- Batch normalization

These methods help prevent overfitting and improve generalization.

E. Model Training

The model used in this project is YOLOv8s, and additional experiments were conducted using YOLOv11 to evaluate performance differences.

The dataset was split into training, validation, and testing sets. The model was trained for multiple epochs using a fixed batch size. During training, validation performance was monitored to prevent overfitting and ensure stable learning.

YOLO was selected because it provides fast and accurate object detection by processing the entire image in a single pass. This makes it suitable for real-time applications.

F. Visualization (EDA)

Exploratory Data Analysis (EDA) was conducted to better understand the dataset before training.

- **Class Distribution:** A bar chart was used to show the number of images per class, ensuring balanced representation.
- **Image Size Analysis:** Histograms were used to analyze image dimensions. Images were later resized to a standard size for training.
- **Bounding Box Analysis:** The distribution of bounding box sizes shows that objects are clearly visible in most images, which supports effective detection.

Overall, the dataset is well-prepared and suitable for training a YOLO-based model. The model used in this project is YOLOv8s, and additional experiments were conducted using YOLOv11 to evaluate performance differences. The dataset was divided into training, validation, and testing sets to ensure proper evaluation. The model was trained for 50 epochs with a batch size of 16, which allowed stable learning and effective convergence. During training, validation performance was continuously monitored to prevent overfitting and ensure that the model generalizes well to unseen data. YOLO was selected because it provides fast and accurate object detection by processing the entire image in a single pass, making it highly suitable for real-time applications.

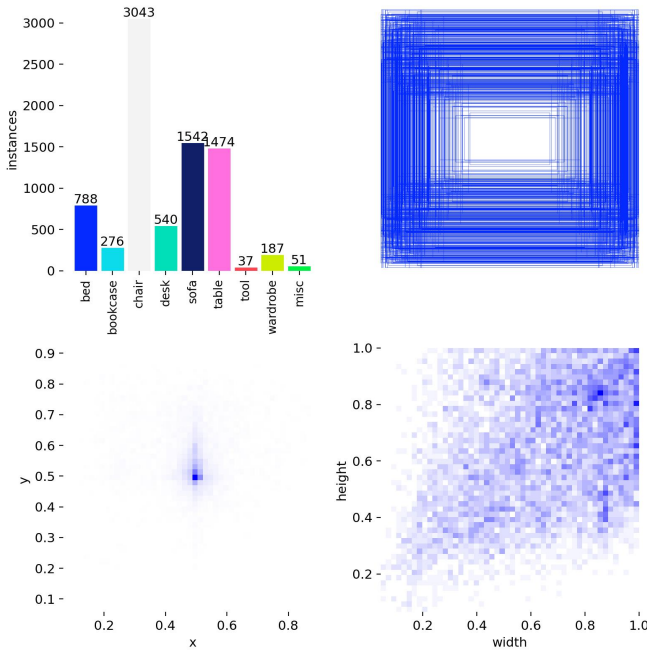


Fig. 1. Furniture dataset analysis: class distribution, spatial density, and object dimensions.

VIII. DISCUSSION AND RESULTS

The YOLO-based model was tested on sample images to evaluate its performance in object detection. The results show that the model can successfully detect multiple objects within a single image and provide structured outputs.

When testing on image `im2.jpg`, the model detected:

- Total objects: 5
- 4 chairs
- 1 table

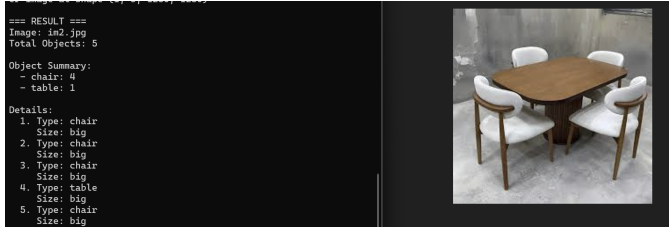


Fig. 2. Example output From our model

These results demonstrate that the model is capable of:

- Detecting multiple objects in a single image
- Accurately counting objects

The system also provides additional details such as object size, which improves the clarity of the output.

In addition to the detection results, the model performance was evaluated using standard object detection metrics. The model achieved a precision of 79.0%, which indicates that most of the detected objects were correctly classified with relatively few false positives. The recall reached 85.7%, showing that the model was able to detect the majority of objects

present in the images. Furthermore, the model achieved a mean Average Precision (mAP@0.5) of 90.4%, reflecting strong overall detection performance at an intersection-over-union threshold of 0.5. These results demonstrate that the model performs effectively in identifying and detecting objects, providing reliable outputs for real-time applications.

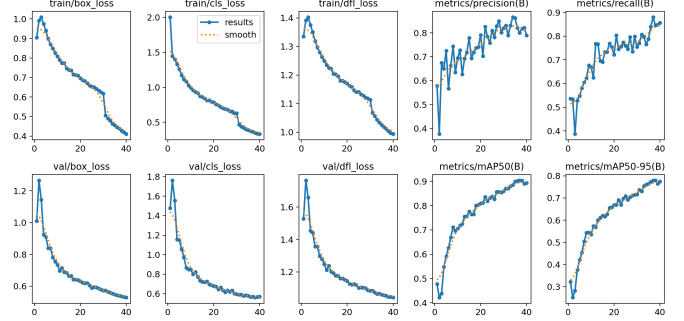


Fig. 3. Training and validation loss trends alongside accuracy metrics over 40 epochs.

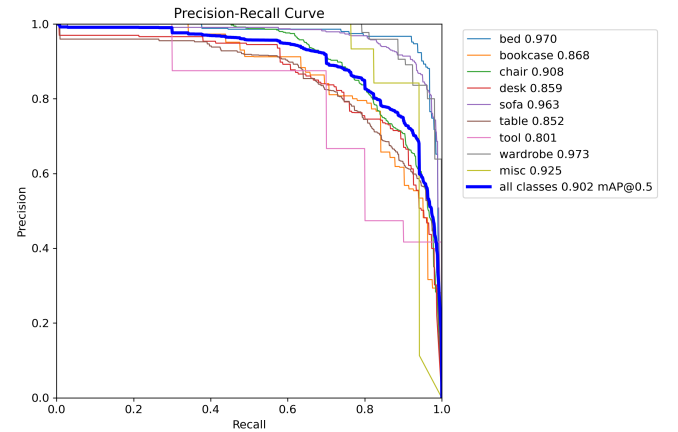


Fig. 4. Per-class Precision-Recall curves and mean Average Precision (mAP@0.5) for the detection model.

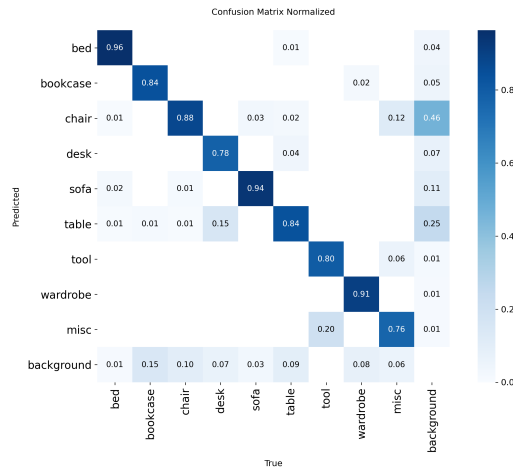


Fig. 5. A visualization of prediction accuracy and classification errors across all furniture categories.

One limitation observed is that all detected objects were classified as “big,” which indicates that size estimation is simple and may need improvement.

Another limitation is that the current model detects object types rather than recyclable materials. This means that although the detection pipeline is working correctly, it still needs to be extended to classify materials such as plastic, metal, and wood.

Overall, the model performs well and provides reliable results. It shows that YOLO is effective for real-time detection tasks. These results confirm that the detection system can be applied to recycling environments by replacing object categories with material classes, enabling automated sorting and improving recycling efficiency.

IX. CONCLUSION

This project presented an object detection system using a YOLO-based model to identify and classify objects from images. The system successfully detects multiple objects, counts them, and provides structured outputs.

The results show that the model performs well in real-time detection and reduces the need for manual identification.

Although the current implementation focuses on object types, it provides a strong foundation for future work. The system can be extended by training the model on recycling-specific datasets to classify materials such as plastic, metal, and wood.

Future improvements may include improving size estimation, increasing dataset diversity, and enhancing model accuracy.

REFERENCES

- [1] Elsevier, “A critical review of plastic waste recycling technology,” *Current Research in Green and Sustainable Chemistry*, 2023.
- [2] R. Geyer, J. R. Jambeck, and K. L. Law, “Production, use, and fate of all plastics ever made,” *Science Advances*, 2020.
- [3] Y. Zhang *et al.*, “Polymer recycling advances,” *PubMed Central*, 2025.

- [4] SAGE Publications, “Waste management study,” *Waste Management & Research*, 2024.
- [5] J. Hopewell, R. Dvorak, and E. Kosior, “Mechanical recycling of packaging plastics,” *Macromolecular Rapid Communications*, 2020.
- [6] Elsevier, “The world of plastic waste: A review,” *Sustainable Chemistry for the Environment*, 2024.
- [7] ACS Publications, “Chemical recycling of plastics,” *ACS Sustainable Chemistry & Engineering*, 2024.
- [8] MDPI, “Plastic recycling methods,” *Sustainability*, vol. 14, no. 24, p. 16340, 2022.
- [9] Nature, “Plastic recycling and circular economy,” *npj Sustainable Manufacturing*, 2024.
- [10] Elsevier, “Solid waste recycling study,” *Waste Management*, 2018.
- [11] Elsevier, “Waste management analysis,” *Waste Management*, 2019.
- [12] Y. LeCun, Y. Bengio, and G. Hinton, “Deep learning,” *Nature*, vol. 521, no. 7553, pp. 436–444, 2015.
- [13] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, “You only look once: Unified, real-time object detection,” arXiv preprint arXiv:1506.02640, 2016.
- [14] A. Bochkovskiy, C.-Y. Wang, and H.-Y. M. Liao, “YOLOv4: Optimal speed and accuracy of object detection,” arXiv preprint arXiv:2004.10934, 2020.
- [15] D. Boragaonkar *et al.*, “Smart waste classification system using YOLO framework,” in *IEEE Conference Proceedings*, IEEE, 2025.
- [16] A. M. Nasir and R. Roslan, “Real-time recycle waste classification using deep learning,” in *IEEE Conference Proceedings*, IEEE, 2024.